

Experiment-8

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Course Code: -MLA0305

Slot-B

Title: - Implementation and Comparative Analysis of SARSA and Q-Learning

Aim- To implement and compare the performance of the SARSA and Q-Learning algorithms for solving a Reinforcement Learning problem.

Procedure

A. SARSA

1. Initialize the Q-table.
2. Select the initial state.
3. Select an action using an ϵ -greedy policy.
4. Execute the action.
5. Observe the reward and next state.
6. Select the next action using the same policy.
7. Update the Q-value using:

$$Q(S,A) \leftarrow Q(S,A) + \alpha [R + \gamma Q(S',A') - Q(S,A)]$$

8. Repeat until the episode terminates.

B. Q-Learning

9. Initialize another Q-table.
10. Select an action using an ϵ -greedy policy.
11. Execute the action and observe the reward and next state.
12. Select the action having the maximum Q-value for the next state.
13. Update the Q-value using:

$$Q(S,A) \leftarrow Q(S,A) + \alpha [R + \gamma \max_{A'} Q(S',A') - Q(S,A)]$$

14. Repeat the process for multiple episodes.
15. Calculate the cumulative reward obtained by both algorithms.
16. Plot the learning curves.
17. Compare SARSA and Q-Learning based on reward, convergence, and

learning performance. **Source Code**

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.mixture import GaussianMixture
from IPython.display import display, FileLink
from google.colab import files
np.random.seed(42)

print("=" * 80)
print("EXPERIMENT 8")
print("SARSA AND Q-LEARNING COMPARATIVE ANALYSIS")
print("=" * 80)
N =
```

```

500 base = np.column_stack([ np.random.randint(0, 10, N),
np.random.randint(0, 2, N),  np.random.randint(0, 10, N),
np.random.uniform(-5, 10, N)
])
gmm = GaussianMixture(
n_components=3,  random_state=42
)
gmm.fit(base)

generated = gmm.sample(N)[0]

dataset = pd.DataFrame(
generated,  columns=[
    "State",
    "Action",
    "Next_State",
    "Reward"
]
)

dataset["State"] = dataset["State"].clip(0, 9).round().astype(int) dataset["Action"]
= dataset["Action"].clip(0, 1).round().astype(int) dataset["Next_State"] =
dataset["Next_State"].clip(0, 9).round().astype(int) dataset["Reward"] =
dataset["Reward"].clip(-5, 10).round(2)

# ===== #
2. 10 SAMPLE RECORDS
# =====

print("\n10 SAMPLE DATASET RECORDS")

display(dataset.head(10))

print("Total Records:", len(dataset))

# ===== #
3. DATASET SUMMARY
# =====

display(
    dataset.describe().T[
        ["min", "max", "mean", "std"]
    ].round(3)
)

# =====
# 4. DOWNLOAD DATASET

```

```

# =====

filename = "Experiment_8_SARSA_Q_Learning_GMM_Dataset.csv"

dataset.to_csv(
    filename, index=False
)

print("\nDOWNLOAD DATASET:")

display(FileLink(filename))

files.download(filename)

# ===== #
5. PARAMETERS
# =====

N_STATES = 10
N_ACTIONS = 2

EPISODES = 500

ALPHA = 0.10
GAMMA = 0.90
EPSILON = 0.10

# =====
# 6. ENVIRONMENT
# =====

def step(state, action):

    if action == 0:

        next_state = min(
            state + 1,
            9
        )

    else:

        next_state = max(
            state - 1,
            0
        )

```

```

        if next_state == 9:

            reward = 10
            done = True

        else:

            reward = -1
            done = False

        return next_state, reward, done

# ===== #
7. ACTION SELECTION
# =====

def choose_action(Q, state):

    if np.random.rand() < EPSILON:

        return np.random.randint(
            N_ACTIONS
        )

    return np.argmax(
        Q[state]
    )

# ===== #
8. SARSA
# =====
Q_sarsa = np.zeros(
    (N_STATES, N_ACTIONS)
)

sarsa_rewards = []

for episode in range(EPISODES):

    state = 0

    action = choose_action(
        Q_sarsa,
        state
    )

```

```
total_reward = 0
done = False

while not done:

    next_state, reward, done = step(
        state,
action
    )

    next_action = choose_action(
        Q_sarsa,
next_state
    )

    if done:

        target = reward

    else:

        target = (
reward +
GAMMA *
Q_sarsa[
next_state,
next_action
        ]
    )
    Q_sarsa[state, action] += (
        ALPHA *
        (
            target -
            Q_sarsa[state, action]
        )
    )

    state = next_state
action = next_action

    total_reward += reward

    sarsa_rewards.append(
total_reward
    )
```

```

# ===== #
9. Q-LEARNING
# =====

Q_qlearning = np.zeros(
    (N_STATES, N_ACTIONS)
)

qlearning_rewards = []

for episode in range(EPISODES):

    state = 0
    total_reward = 0    done
    = False

    while not done:

        action = choose_action(
            Q_qlearning,    state
        )

        next_state, reward, done = step(
            state,
            action
        )
        if done:

            target = reward

        else:

            target = (
                reward +
                GAMMA *
                np.max(
                    Q_qlearning[next_state]
                )
            )

        Q_qlearning[state, action] += (
            ALPHA *
            (
                target -
                Q_qlearning[state, action]
            )
        )

```

```

        state = next_state

        total_reward += reward

        qlearning_rewards.append(
total_reward
        )

# ===== #
10. Q-TABLE RESULTS
# =====

sarsa_table = pd.DataFrame(
    Q_sarsa,
    columns=[
        "Action 0 - Forward",
        "Action 1 - Backward"
    ]
)

sarsa_table.insert(
    0,
    "State",
    range(N_STATES)
)

qlearning_table = pd.DataFrame(
    Q_qlearning, columns=[
        "Action 0 - Forward",
        "Action 1 - Backward"
    ]
)

qlearning_table.insert(
    0,
    "State",
    range(N_STATES)
)

print("\nSARSA Q-TABLE")

display(
    sarsa_table.round(3)
)

print("\nQ-LEARNING Q-TABLE")

```

```

display(
qlearning_table.round(3)
)

# =====
# 11. COMPARISON TABLE
# =====

comparison = pd.DataFrame({
    "Algorithm": [
        "SARSA",
        "Q-Learning"
    ],

    "Total Reward": [
np.sum(sarsa_rewards),
np.sum(qlearning_rewards)
    ],

    "Average Reward": [
np.mean(sarsa_rewards),
np.mean(qlearning_rewards)
    ],

    "Maximum Q Value": [
np.max(Q_sarsa),
np.max(Q_qlearning)
    ],

    "Best Action State 0": [
np.argmax(Q_sarsa[0]),
np.argmax(Q_qlearning[0])
    ]
}).round(3)

print("\n" + "=" * 80) print("SARSA VS Q-
LEARNING COMPARISON") print("=" * 80)

display(comparison)

# =====
# 12. SUMMARY TABLE
# =====

best_algorithm = comparison.loc[
comparison["Average Reward"].idxmax(),

```



```

        "Algorithm"
    ]

summary = pd.DataFrame({
    "Metric": [
        "Dataset Records",
        "Episodes",
        "States",
        "Actions",
        "Learning Rate",
        "Discount Factor",
        "Epsilon",
        "SARSA Average Reward",
        "Q-Learning Average Reward",
        "Best Algorithm"
    ],

    "Result": [
len(dataset),
EPISODES,
    N_STATES,
    N_ACTIONS,
    ALPHA,
    GAMMA,          EPSILON,
round(np.mean(sarsa_rewards),    3),
round(np.mean(qlearning_rewards), 3),
best_algorithm
    ]
})

print("\nSUMMARY RESULTS")

display(summary)

# =====
# 13. VISUALIZATION 1
# DATASET REWARD
# =====

plt.figure(figsize=(10, 6))

plt.hist(
dataset["Reward"],
bins=25,
color="#845ef7",
edgecolor="black"
)

```

```

plt.title(
    "GMM Synthetic Dataset - Reward Distribution",
    fontsize=16, fontweight="bold"
)

plt.xlabel("Reward") plt.ylabel("Frequency")

plt.show()

# =====
# 14. VISUALIZATION 2
# RAW EPISODE REWARDS
# =====

plt.figure(figsize=(11, 6))

plt.plot(
    sarsa_rewards,
    color="#ff6b6b",
    alpha=0.6,
    label="SARSA"
)

plt.plot(
    qlearning_rewards,
    color="#339af0",
    alpha=0.6, label="Q-
Learning"
)

plt.title(
    "SARSA vs Q-Learning Episode Rewards",
    fontsize=16, fontweight="bold"
)

plt.xlabel("Episode") plt.ylabel("Reward")

plt.legend() plt.grid(alpha=0.25)

plt.show()

# =====
# 15. VISUALIZATION 3
# MOVING AVERAGE
# =====

```

```
window = 25

sarsa_ma = pd.Series(
    sarsa_rewards
).rolling(window).mean()

qlearning_ma = pd.Series(
    qlearning_rewards
).rolling(window).mean()

plt.figure(figsize=(11, 6))

plt.plot(
    sarsa_ma,
    color="#e03131",
    linewidth=3,
    label="SARSA"
)

plt.plot(
    qlearning_ma,
    color="#1971c2",
    linewidth=3,
    label="Q-Learning"
)

plt.title(
    "SARSA vs Q-Learning Moving Average Reward",
    fontsize=16, fontweight="bold"
)

plt.xlabel("Episode") plt.ylabel("25-Episode
Average Reward")

plt.legend() plt.grid(alpha=0.25)

plt.show() plt.figure(figsize=(9,
6))

plt.bar(
    ["SARSA", "Q-Learning"],
    [
        np.max(Q_sarsa),
        np.max(Q_qlearning)
    ],
    color=["#ff6b6b", "#339af0"],
    edgecolor="black"
```

```

) plt.title(
    "Maximum Q-Value Comparison",
    fontsize=16,
    fontweight="bold"
)
plt.ylabel("Maximum Q Value")
plt.show() fig, axes =
plt.subplots(
    1,
    2,
    figsize=(15, 7)
)
im1 = axes[0].imshow(
    Q_sarsa,
    cmap="turbo",
    aspect="auto"
) axes[0].set_title(
    "SARSA Q-Table",
    fontsize=15,
    fontweight="bold"
)
axes[0].set_xlabel("Action") axes[0].set_ylabel("State")

axes[0].set_xticks(
    [0, 1]
)
axes[0].set_xticklabels(
    ["Forward", "Backward"]
)
axes[0].set_yticks(
    range(N_STATES)
)
fig.colorbar(
    im1,
    ax=axes[0]
)
im2 = axes[1].imshow(
    Q_qlearning,
    cmap="plasma",
    aspect="auto"
) axes[1].set_title(
    "Q-
Learning Q-Table",
    fontsize=15,
    fontweight="bold"
)
axes[1].set_xlabel("Action") axes[1].set_ylabel("State")

```

```

axes[1].set_xticks(
    [0, 1]
)
axes[1].set_xticklabels(
    ["Forward", "Backward"]
)
axes[1].set_yticks(
    range(N_STATES)
)
fig.colorbar(
    im2,
    ax=axes[1]
)
plt.tight_layout()
plt.show() print("\n" +
"=" * 80) print("FINAL
RESULT") print("=" *
80)

print(
    f"""
SARSA Average Reward    : {np.mean(sarsa_rewards):.3f}
Q-Learning Average Reward : {np.mean(qlearning_rewards):.3f}
Better performing algorithm based on average reward:
{best_algorithm} 500 GMM records generated.
10 sample records displayed.
Complete CSV dataset available for download.
"""
)
print("Experiment 8 completed successfully.")

```

State	Action	Next_State	Reward	
0	5	1	0	5.38
1	6	0	3	7.05
2	4	0	5	0.97
3	8	0	3	6.24
4	0	0	3	5.83
5	7	1	4	5.87
6	7	1	3	4.90
7	4	0	1	3.29
8	4	1	4	0.88
9	3	0	3	6.57

FIG: - 1 SAMPLE DATA

	min	max	mean	std
State	0.0	9.0	4.494	2.552
Action	0.0	1.0	0.486	0.500
Next_State	0.0	9.0	4.600	2.767
Reward	-5.0	10.0	2.527	4.051

FIG: - 2 MEAN & SD

State	Action 0 - Forward	Action 1 - Backward
0	0	-2.027
1	1	-1.108
2	2	-0.047
3	3	1.178
4	4	2.606
5	5	3.836
6	6	5.532
7	7	7.890
8	8	10.000
9	9	0.000

State	Action 0 - Forward	Action 1 - Backward
0	0	-1.391
1	1	-0.434
2	2	0.629
3	3	1.810
4	4	3.122
5	5	4.580
6	6	6.200
7	7	8.000
8	8	10.000
9	9	0.000

FIG: - 3 SARSA Q-TABLE vs Q-LEARNING Q-TABLE

	Metric	Result
0	Dataset Records	500
1	Episodes	500
2	States	10
3	Actions	2
4	Learning Rate	0.1
5	Discount Factor	0.9
6	Epsilon	0.1
7	SARSA Average Reward	0.4
8	Q-Learning Average Reward	0.498
9	Best Algorithm	Q-Learning

FIG: - 4 SUMMARY TABLE

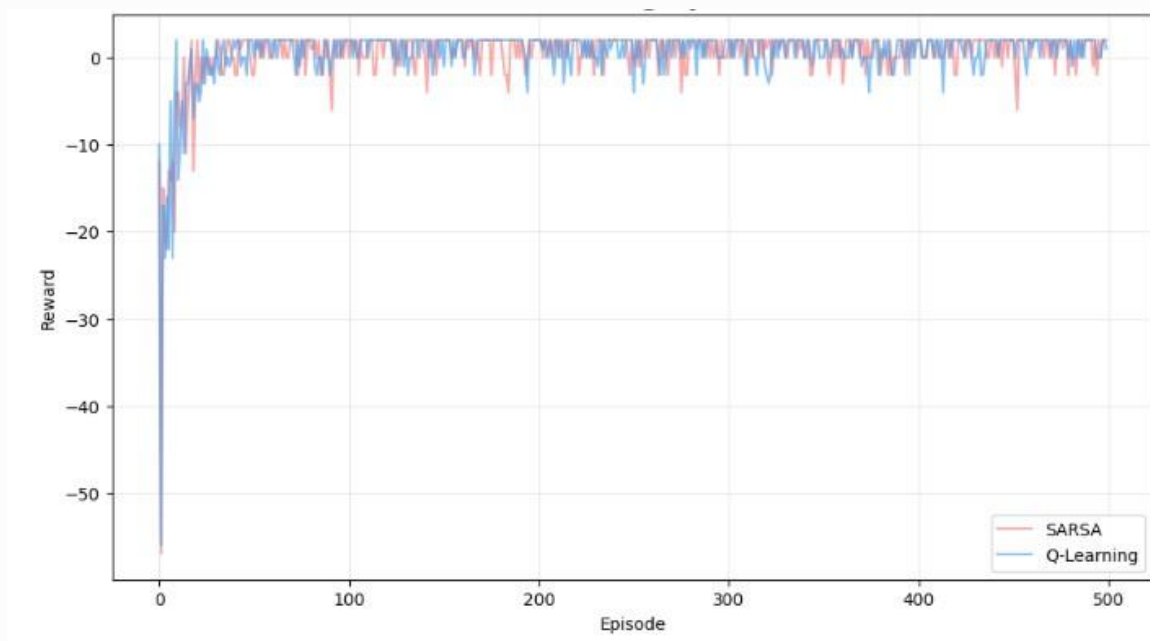


FIG: - 5 SARSA vs Q-LEARNING

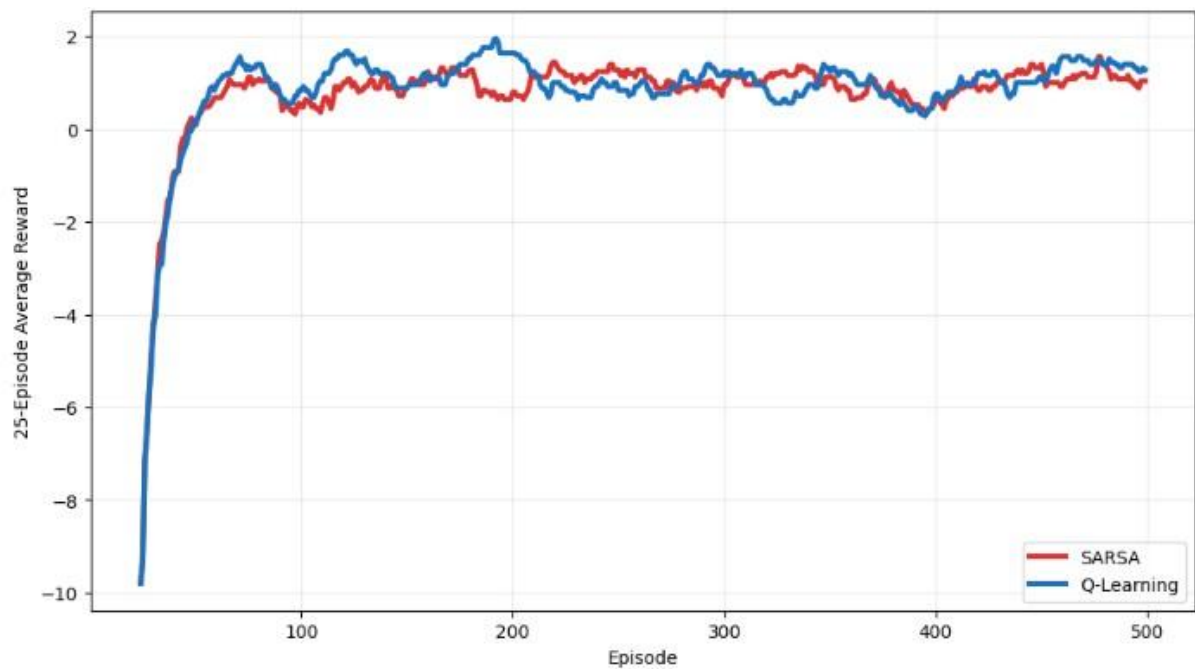


FIG: - 6 SARSA vs Q LEARNING AVG REWARD

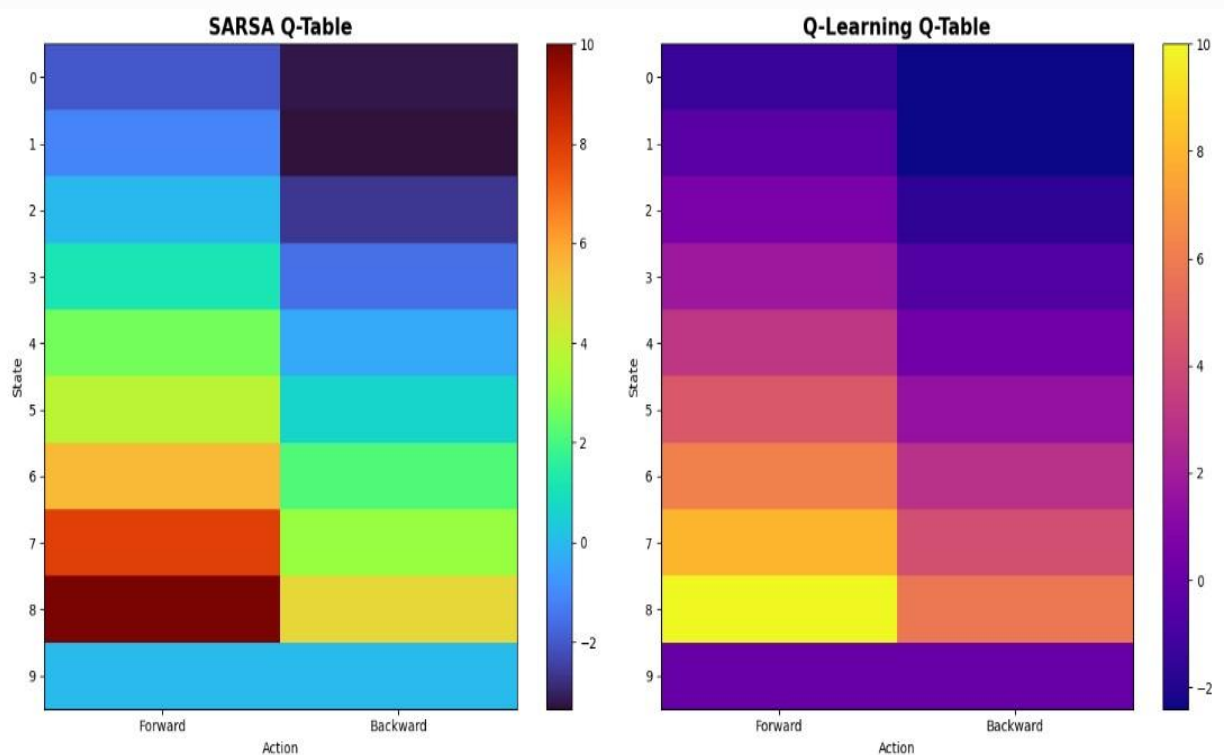
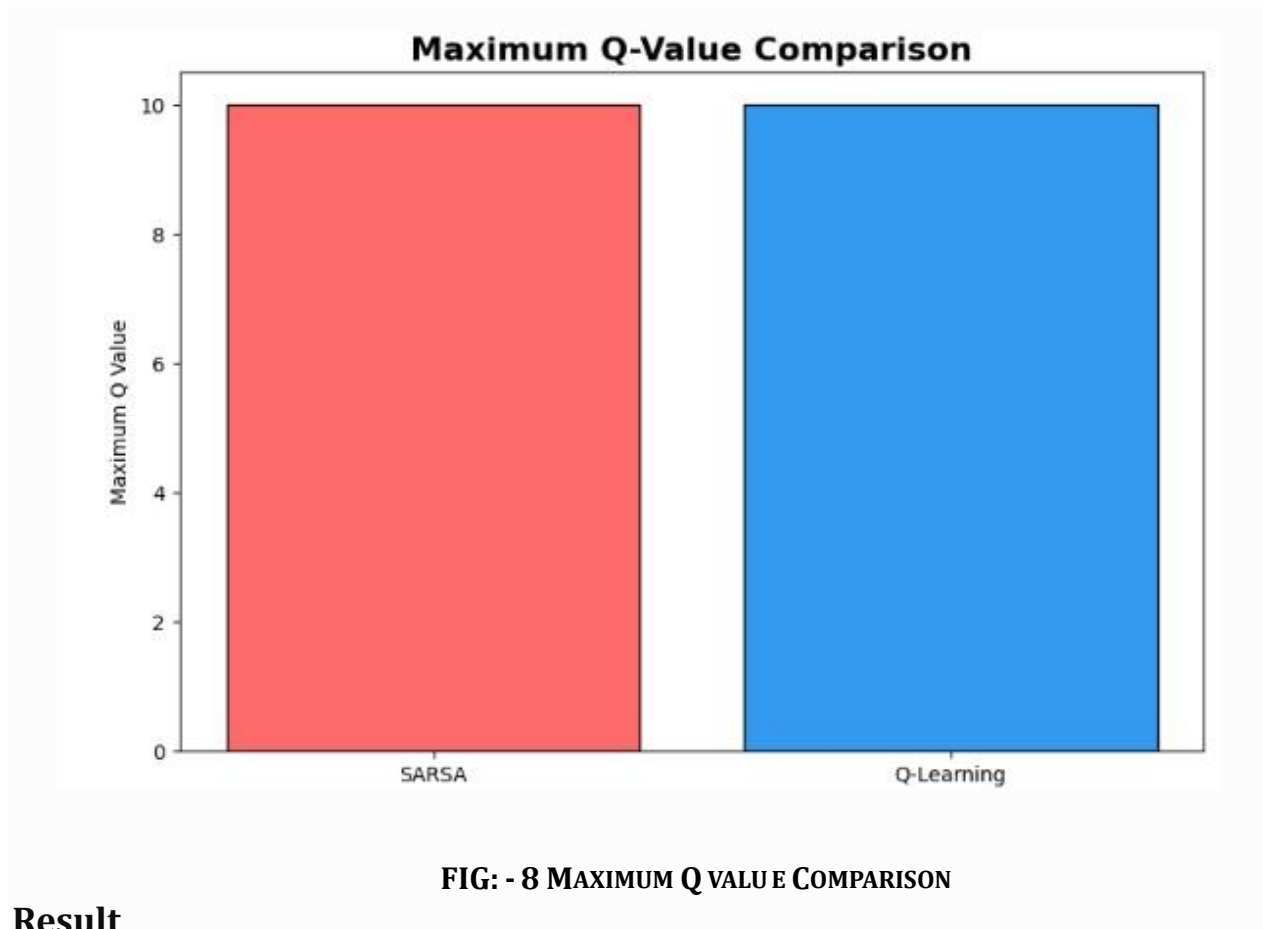


FIG: - 7 VALUE ESTIMATION



Result

Monte Carlo Prediction and Control were successfully implemented. The agent learned from complete episodes and gradually improved its policy based on the estimated returns.